Kalman Filter on IMU Linear Acceleration Integration.

Current Code: (Linear acceleration bias filter + velocity corrector by filtering acceleration bias)

state $x = \begin{bmatrix} V_I^W \\ b_a^B \end{bmatrix} = \begin{bmatrix} \text{vel of body in world frame} \\ \text{bias of body linear acceleration in body frame} \end{bmatrix}$.

at start up: $a_{true}^B = 0$, $V^W = 0$. read q_{WB}, w, α

measuremene: $a_m^B = b_a^B + n_a n_a$ $\quad q_{WB} = q_{WI}$

since $Z_k = H_k X_k + n_k$

$X_k = \begin{bmatrix} V_{Ik}^W \\ b_{a_k}^B \end{bmatrix} \rightarrow b_a^B = H_k \cdot X_k = [0\ I] \cdot \begin{bmatrix} V_{Ik}^W \\ b_{a_k}^B \end{bmatrix} = b_{a_k}^B$.

So $\boxed{H_k = [0\ I].}$, $\boxed{Z_k = a_m^B = H_k \cdot X_k + n_{a_k}.}$ (acceleration measurement noise.)

~~then~~ to obtain initial bias estimate:

$$E[a_m^B] = E[b_a^B] + E[n_a n_a] = E[b_a^B].$$ ($E[n_a n_a] \rightarrow 0$ from Gaussian Distribution.)

if use mean bias as bias estimate prediction:

then $\boxed{\hat{b}_{a_k}^B = E[a_{m_{k-1}}^B] = \frac{1}{K-1} \sum_{i=1}^{K-1} a_{m_i}^B}$ (start up loop reading)

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after initialized, read $a_{m_1}^B$

$a_{I\text{corrected}_k}^B = a_{m_{k-1}}^B - \hat{b}_{a_{k-1}}^{B+}$

after reads q_{WI}.

$q_{WI} \rightarrow R_{WI} = R_{WB}$

$\hat{a}_{I,k-1}^W = R_{WB_{k-1}} \cdot \hat{a}_{I,k-1}^B$

$\hat{V}_{I,k}^W = \hat{V}_{I,k-1}^W + \hat{a}_{I,k-1}^W \cdot \Delta t$.

$\hat{b}_{a,k-1}^{B,-} = \hat{b}_{a,k-2}^{B+}$

$\hat{X}_k^- = \begin{bmatrix} \hat{V}_{I,k}^{W-} \\ \hat{b}_{a,k-1}^{B-} \end{bmatrix}$

also initialize covariance

$P_0 = \begin{bmatrix} P_{VV} & 0 \\ 0 & P_{bb} \end{bmatrix}$.

$P_{VV} = E[VV^T]$

since

$$\begin{cases} \hat{V}_{I,k}^{w,-} = I \cdot \hat{V}_{I,k-1}^{w+} + \hat{a}_{I,k-1}^{w} \cdot \Delta t = I \cdot \hat{V}_{I,k-1}^{w+} + R_{wB_{k-1}} \hat{a}_{I,k-1}^{B} \cdot \Delta t \\ \hat{a}_{I,k}^{w} = R_{wB_k} \hat{a}_{I,k}^{B} \\ \hat{b}_{a,k-1}^{B-} = \hat{b}_{a,k-2}^{B+} \end{cases}$$

scalar

$$= I \cdot \hat{V}_{I,k-1}^{w+} + R_{wB_{k-1}} \cdot \Delta t \; \hat{a}_{I,k-1}^{B}$$

for state $\hat{x}_k^- = \begin{bmatrix} \hat{V}_{I,k}^{w-} \\ \hat{b}_{a,k-1}^{B-} \end{bmatrix} = \begin{bmatrix} I \; \hat{V}_{I,k-1}^{w+} + R_{wB_{k-1}} \cdot \hat{a}_{I,k-1}^{B} \cdot \Delta t \\ \hat{b}_{a,k-2}^{B+} \end{bmatrix}$.

$$\hat{x}_k^- = \begin{bmatrix} I \cdot \hat{V}_{I,k-1}^{w,+} + R_{wB_{k-1}} \left(a_{m,k-1}^{B} - \hat{b}a_{k-1}^{B} \right) \Delta t \\ I \; \hat{b}_{a,k-2}^{B+} \end{bmatrix}$$

$$\hat{x}_k^- = \begin{bmatrix} I \cdot \hat{V}_{I,k-1}^{w+} \\ I \cdot \hat{b}_{a,k-2}^{B,+} \end{bmatrix} + \begin{bmatrix} R_{wB_{k-1}} (a_{m,k-1}^{B}) \Delta t - (R_{wB_{k-1}} \cdot \hat{b}a_{k-1}^{B}) \Delta t \\ 0 \end{bmatrix}$$

$$\hat{x}_k^- = \begin{bmatrix} I \cdot \hat{V}_{I,k-1}^{w,+} \\ I \cdot \hat{b}_{a,k-2}^{B,+} \end{bmatrix} + \begin{bmatrix} -R_{wB_{k-1}} \cdot \Delta t \\ 0 \end{bmatrix} \hat{b}a_{k-2}^{B} + \begin{bmatrix} R_{wB_{k-1}} \Delta t \\ 0 \end{bmatrix} a_{m,k-1}^{B}.$$

$$\hat{x}_k^- = \left(\begin{bmatrix} I & -R_{wB_{k-1}} \cdot \Delta t \\ 0 & I \end{bmatrix} \right) \begin{bmatrix} \hat{V}_{I,k-1}^{w+} \\ \hat{b}_{a,k-2}^{B} \end{bmatrix} + \begin{bmatrix} R_{wB_{k-1}} \Delta t \\ 0 \end{bmatrix} a_{m,k-1}^{B}.$$

F_{k-1} $\quad$ $\hat{x}_{k-1}$ $\quad$ B_{k-1} $\quad$ u_{k-1}

then $P_k^- = F_k \cdot P_{k-1}^+ \cdot F_k^T + Q_k$.

$$P_{k-1} = \begin{bmatrix} P_{k-1}(V_{k-1} V_{k-1}^T) & 0 \\ 0 & P_{k-1}(b_{k-1} b_{k-1}^T) \end{bmatrix}.$$

$$z_k = H_k \cdot x_k + n_{a_k}, \quad H_k = [0, I].$$

$$a_{m_{Ik}}^B = b_{a_k}^B + n_{a_k} \Rightarrow a_{m,k}^B = [0, I] \cdot \begin{bmatrix} v_{I,k}^w \\ b_{a,k}^B \end{bmatrix} + n_{a_k}.$$

essentially are using kalman Filter to filter Linear acceleration bias at stationary, and use last filtered bias to integrate into current linear velocity.

$$S_k = H_k \cdot P_k^- H_k^T + R_k$$

$$K_k = P_k^- H_k^T \cdot S_k^{-T}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k y_k.$$

$$P_k^+ = (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T.$$

$$y_k = a_{m,k}^B - H_k \hat{x}_k^-, \quad \hat{x}_k^- = F_{k-1} \hat{x}_{k-1} + B_{k-1} \cdot u_{k-1}$$

$$= \begin{bmatrix} \hat{v}_{I,k-1}^{w,-} \\ \hat{b}_{a,k-1}^{B,-} \end{bmatrix} \begin{bmatrix} \hat{v}_{I,k}^{w,+} \\ \hat{b}_{a,k-1}^{B,-} \end{bmatrix}$$

$$F_{k-1} = \begin{bmatrix} I & -R_{k-1}^{wB} \Delta t \\ 0 & I \end{bmatrix}$$

$$B_{k-1} = \begin{bmatrix} R_{k-1}^{wB} \Delta t \\ 0 \end{bmatrix}$$

$$\hat{x}_k^+ = \begin{bmatrix} \hat{v}_k^{w,+} \\ \hat{b}_{a,k-1}^{B,+} \end{bmatrix} \rightarrow \begin{cases} a_{I\,\text{correceed}}^B = a_{m_k}^B - \hat{b}_{a,k-1}^{B,+} \\ \hat{v}_{w_k}^{w,+} \end{cases}$$